Graph Theory – Revision Notes

Discrete Mathematics — BCA Honours

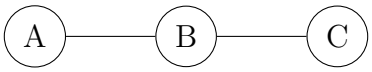
Based on Kenneth H. Rosen's *Discrete Mathematics and Its Applications with Combinatorics and Graph Theory*

1 Introduction to Graphs

A graph is a pictorial representation of relationships between objects. It helps us model connections, such as:

- Friendships in social networks
- Routes between cities
- Flight connections
- Network topologies in computers

Example: Consider 3 cities: A, B, and C. There are roads between A–B and B–C. We can represent this as:



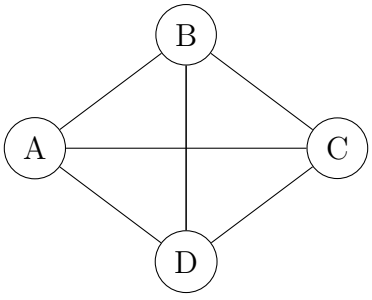
2 The Königsberg Bridge Problem

In the 18th century, the city of Königsberg (now Kaliningrad) had seven bridges connecting different parts of land separated by a river. People wondered:

"Is it possible to take a walk that crosses each bridge exactly once and return to the starting point?"

Euler's Insight: Represent the land areas as vertices and the bridges as edges, forming a graph. He showed that such a walk was not possible, laying the foundation of graph theory.

Graph Representation:



3 Formal Definitions (As per Rosen)

Graph:

A graph G is an ordered pair $G = (V, E)$, where V is a non-empty set of vertices and E is a set of edges (unordered or ordered pairs from V).

Finite and Infinite Graph:

A finite graph has a finite number of vertices and edges. An infinite graph has an infinite number of vertices and/or edges.

Simple Graph:

An undirected graph with no loops and no multiple edges between the same pair of vertices.

Multigraph:

An undirected graph that may have multiple edges between the same pair of vertices, but no loops.

Pseudograph:

A graph that may contain both multiple edges and loops.

Undirected Graph:

A graph where the edges are unordered pairs of vertices (i.e., edge $\{u, v\} = \{v, u\}$).

Directed Graph:

A graph where edges are ordered pairs (arcs), i.e., (u, v) is directed from u to v.

Mixed Graph:

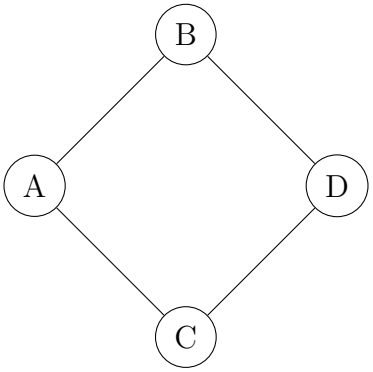
A graph that contains both directed and undirected edges.

Example

Problem: Draw a graph that models the following situation: Let A, B, C, D be people. A is friends with B and C. B is also friends with D. C is friends with D. Draw an undirected graph representing this.

Vertices: A, B, C, D

Edges: {A, B}, {A, C}, {B, D}, {C, D}



Recap Table

Term	Meaning
Graph	Set of vertices and edges
Simple Graph	No loops or multiple edges
Multigraph	Multiple edges, no loops
Pseudograph	Allows both multiple edges and loops
Directed Graph	Edges have direction (arrows)
Undirected Graph	Edges are bidirectional
Mixed Graph	Has both directed and undirected edges
Finite Graph	Limited vertices and edges
Infinite Graph	Unbounded vertices and/or edges

Exercise

Air Pollution Monitoring Network

Five cities have sensors installed that send data to each other for pollution tracking:

- Kochi shares data with Thrissur and Palakkad.
- Palakkad forwards data to Coimbatore.
- Coimbatore shares data with Madurai.
- Madurai sends data back to Kochi once daily.

Model the communication system as:

a) A directed graph showing flow of data.
b) A multigraph if data is sent multiple times per day.
c) A simple graph if we only care whether communication exists.
d) A pseudograph if a city processes and returns data to itself.

4 Basic Terms in Graph Theory

Loop: A loop is an edge that connects a vertex to itself.

Multiple Edges: Multiple edges are two or more edges that connect the same pair of vertices.

Endpoints of an edge: The endpoints of an edge are the two vertices that an edge connects.

Adjacency: Two vertices u and v in an undirected graph G are said to be adjacent if there is an edge $e = \{u, v\}$ in G that connects u and v.

Incidence: An edge e is said to be incident with a vertex v if v is one of the endpoints of e.

Degree: The degree of a vertex v, denoted $\deg(v)$, is the number of edges incident with v in an undirected graph.

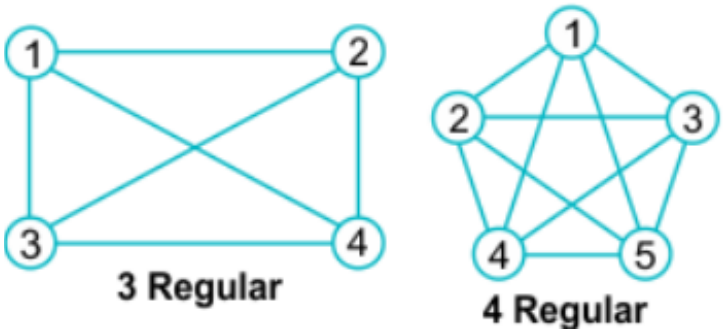
Isolated Vertex: A vertex with degree zero (not connected to any other vertex) is called an isolated vertex.

Pendant Vertex: A vertex with degree one is called a pendant vertex.

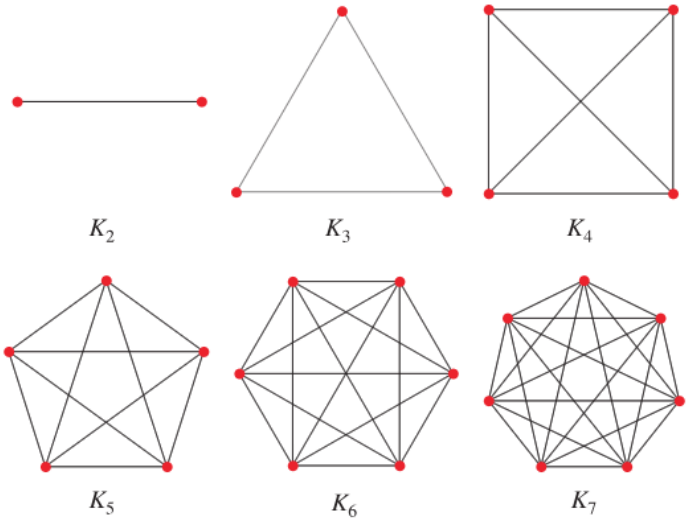
Initial and Terminal Vertex (Directed Graph): In a directed graph, for an edge (u, v), u is the initial vertex and v is the terminal vertex.

In-degree and Out-degree: In a directed graph, the in-degree of a vertex is the number of edges coming into the vertex, and the out-degree is the number of edges going out of the vertex.

Regular Graph. A *regular graph* is a simple graph in which every vertex has the same degree. If every vertex has degree r, then the graph is called an *r-regular graph.*

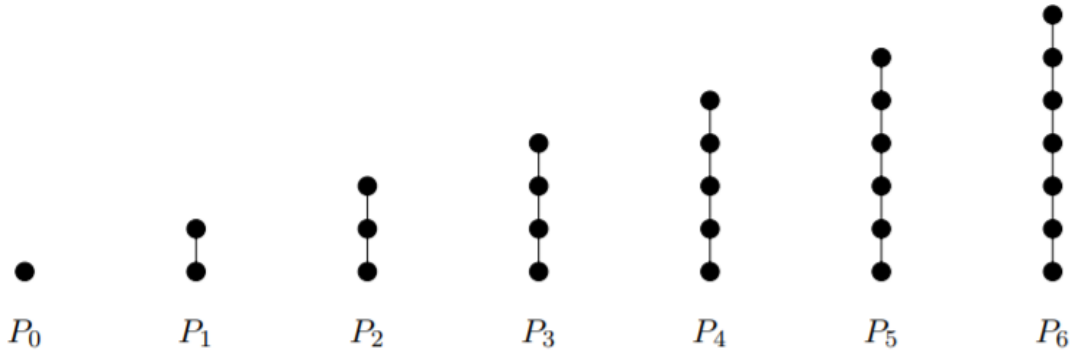


Complete Graph: A complete graph K_n is a simple undirected graph in which every pair of distinct vertices is connected by a unique edge.

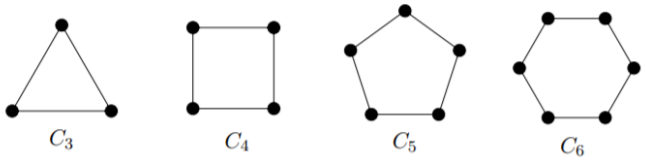


Path: A path P_n is a simple graph consisting of n vertices $v_1, v_2, \ldots, v_n$, and $n-1$ edges of the form

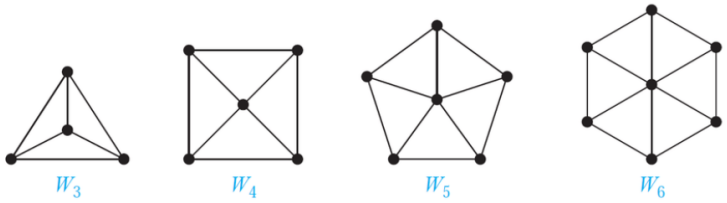
$$\{v_1, v_2\}, \{v_2, v_3\}, \ldots, \{v_{n-1}, v_n\}.$$



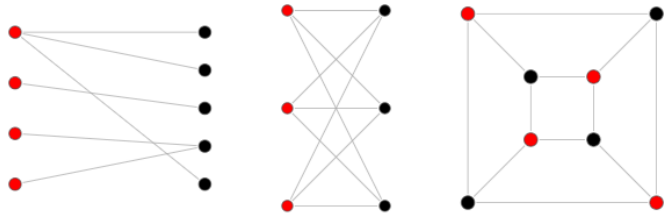
Cycle: A cycle is a closed path in which all edges and all intermediate vertices are distinct.



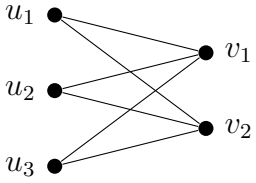
Wheel Graph: A wheel graph W_n is formed by connecting a single vertex to all vertices of a cycle C_n.



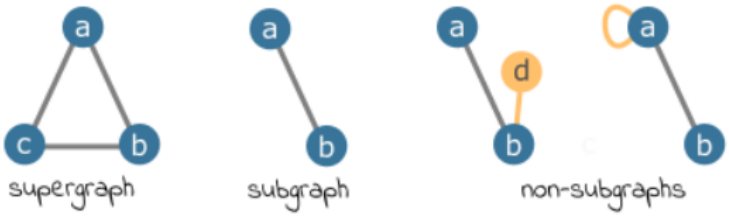
Bipartite Graph: A graph is bipartite if its vertex set can be divided into two disjoint sets V_1 and V_2 such that every edge connects a vertex in V_1 to one in V_2.



Complete Bipartite Graph: A complete bipartite graph $K_{m,n}$ is a bipartite graph where every vertex in V_1 is connected to every vertex in V_2.

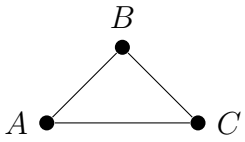


Subgraph: A subgraph of a graph $G = (V, E)$ is a graph $H = (V', E')$ such that $V' \subseteq V$ and $E' \subseteq E$.



Exercise

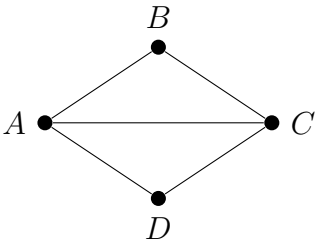
Question: Is the following graph bipartite?



Answer: No, it is not bipartite. A cycle of length 3 (odd length) cannot be bipartite because we cannot split the vertices into two disjoint sets without connecting vertices within the same set.

5 Handshaking Lemma

Statement: In any undirected graph, the sum of the degrees of all vertices is equal to twice the number of edges.



Example:
Vertex degrees: A = 2, B = 2, C = 2, D = 2. Total = 8.
Number of edges = 4.
Hence, $2|E| = \sum \deg(v) = 8$.

Proof: Each edge in an undirected graph contributes exactly 2 to the total degree count (one for each endpoint), so the sum of degrees equals $2|E|$.

Theorem: Even Number of Odd Degree Vertices

Statement: In any undirected graph, the number of vertices with odd degree is even.

Proof: From the handshaking lemma, we know the total sum of all vertex degrees is even. Suppose there were an odd number of vertices with odd degree; their sum would be odd, which contradicts the total being even. Hence, the number of odd-degree vertices must be even.

6 Representation of Graphs

Graphs can be represented in several ways. The most commonly used representations are:

6.1 Adjacency List

An adjacency list represents a graph as an array of lists. The index of the array represents the vertex and each element in its list represents the vertices adjacent to it.

Example: Consider a graph with vertex set $V = \{a, b, c, d\}$ and edges $\{(a, b), (a, d), (b, c), (c, d)\}$.
Adjacency List:

- a: b, d
- b: a, c
- c: b, d
- d: a, c

6.2 Adjacency Matrix

An adjacency matrix is a square matrix used to represent a finite graph. The elements of the matrix indicate whether pairs of vertices are adjacent or not in the graph.

Example:

	a	b	c	d
a	0	1	0	1
b	1	0	1	0
c	0	1	0	1
d	1	0	1	0

6.3 Incidence Matrix

An incidence matrix is a matrix of size $n \times m$ where n is the number of vertices and m is the number of edges. Entry (i, j) is 1 if vertex i is incident to edge j; otherwise, it is 0.

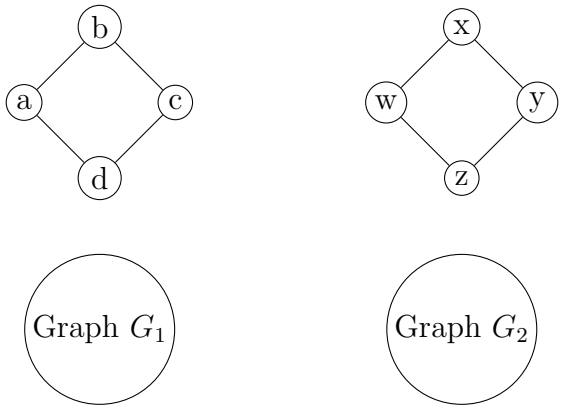
Example: Let the graph have edges $e_1 = (a, b)$, $e_2 = (a, d)$, $e_3 = (b, c)$, $e_4 = (c, d)$.

	e_1	e_2	e_3	e_4
a	1	1	0	0
b	1	0	1	0
c	0	0	1	1
d	0	1	0	1

7 Graph Isomorphism

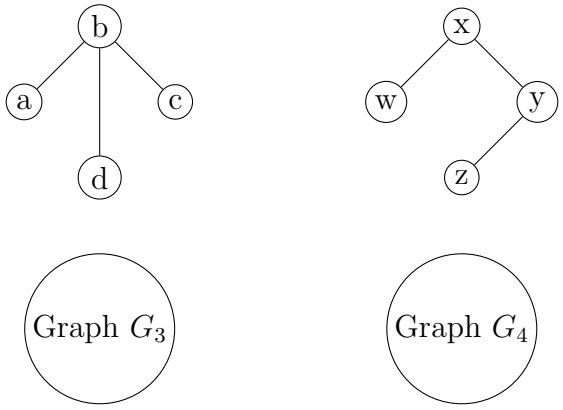
Two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are said to be **isomorphic** if there is a bijection $f : V_1 \to V_2$ such that $\{u, v\} \in E_1$ if and only if $\{f(u), f(v)\} \in E_2$.

Examples of Isomorphic Graphs



Graphs G_1 and G_2 are isomorphic via the mapping $a \to w$, $b \to x$, $c \to y$, $d \to z$.

Non-Isomorphic Graphs (Different Degree Sequence)



Graphs G_3 and G_4 are **not isomorphic** because they have different degree sequences.

8 Connectivity

8.1 Walk

A **walk** in a graph is a sequence of vertices such that each consecutive pair of vertices is connected by an edge in the graph. Vertices and edges may be repeated.

8.2 Closed Walk

A **closed walk** is a walk that starts and ends at the same vertex.

8.3 Path

A **path** is a walk in which no vertex is repeated (except possibly the first and last if it's a closed path).

8.4 Trail

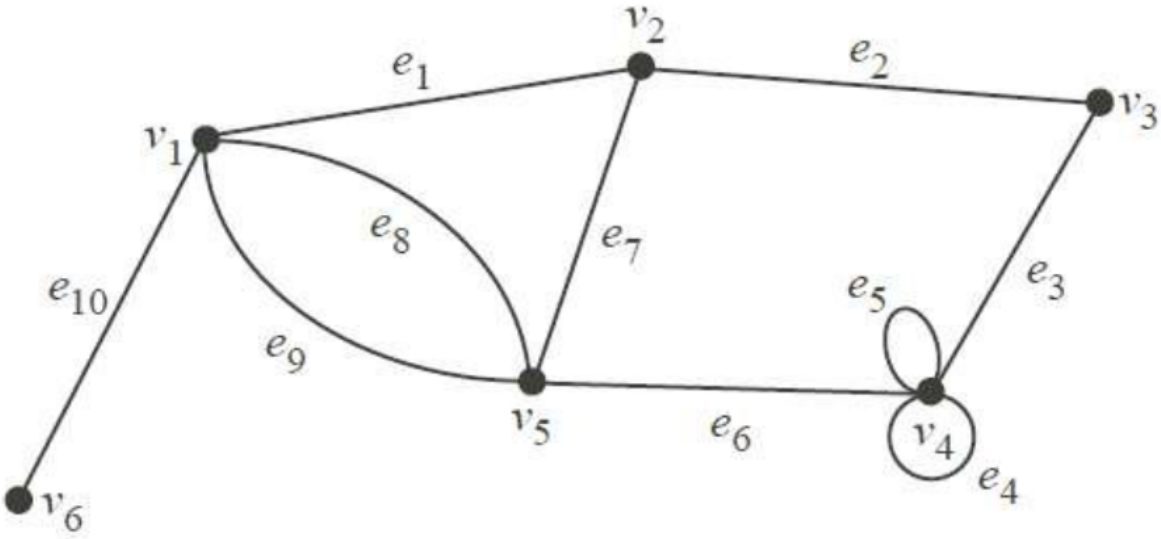
A **trail** is a walk in which all the edges are distinct. Or in other words, a **trail** is a walk in which no edge is repeated (but vertices may be repeated).

8.5 Circuit

A **circuit** is a closed trail — it starts and ends at the same vertex, and no edge is repeated.

8.6 Connected Graph

A graph is said to be **connected** if there is a path between every pair of vertices.



Graph G

Open Walk: $v_2, e_7, v_5, e_8, v_1, e_8, v_5, e_6, v_4, e_5, v_4, e_5, v_4$

Closed Walk: $v_4, e_5, v_4, e_3, v_3, e_2, v_2, e_7, v_5, e_6, v_4$

Trail: $v_1, e_8, v_5, e_9, v_1, e_1, v_2, e_7, v_5, e_6, v_4, e_5, v_4, e_4, v_4$

Path: $v_2, e_7, v_5, e_6, v_4, e_3, v_3$

Circuit: $v_2, e_7, v_5, e_6, v_4, e_3, v_3, e_2, v_2$

8.7 Euler Trail and Circuits

Euler Path: A trail that includes every edge of the graph exactly once.
Euler Circuit: An Euler Trail that is closed — starts and ends at the same vertex, using every edge exactly once.

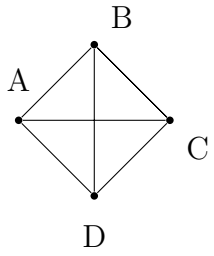
Euler's Theorem:

- A connected graph has an Euler circuit if and only if every vertex has even degree.
- A connected graph has an Euler path (but not a circuit) if and only if exactly two vertices have odd degree.

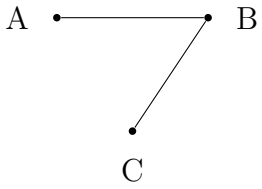
8.8 Hamiltonian Paths and Circuits

Hamiltonian Path: A path that visits every vertex exactly once.
Hamiltonian Circuit: A Hamiltonian path that starts and ends at the same vertex.

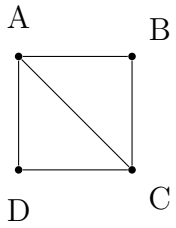
Eulerian Example:



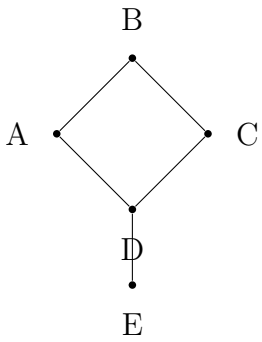
Non-Eulerian Example:

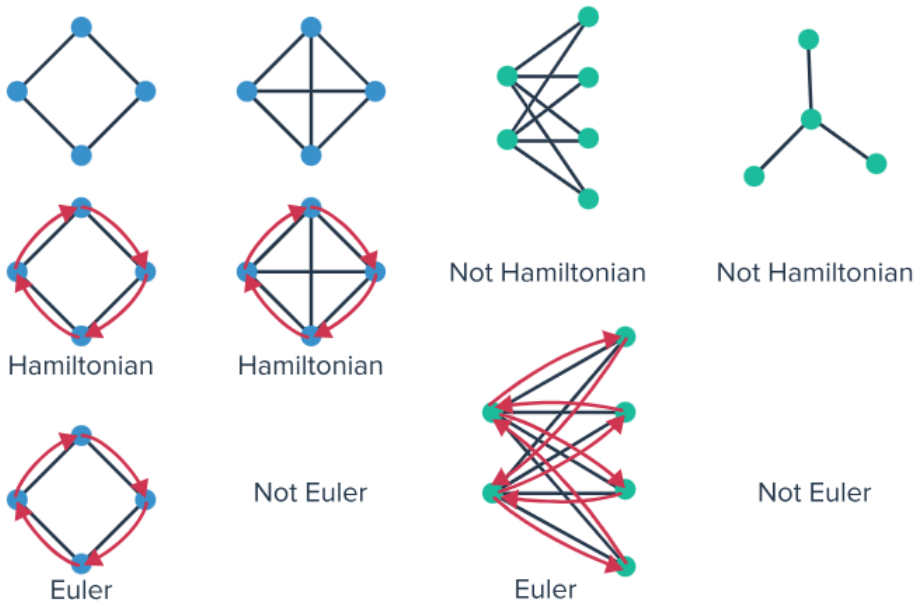


Example (Hamiltonian Circuit):



Counter Example (No Hamiltonian Circuit):





9 Algorithms

9.1 Dijkstra's Algorithm

Definition: Dijkstra's algorithm finds the shortest path from a starting vertex to all other vertices in a weighted graph with non-negative weights.

Algorithm Steps

1. Construct a table with columns corresponding to every vertex of the graph.
2. In the first row, assign the value 0 to the source vertex and ∞ (infinity) to all other vertices.
3. To fill the next row, look at all vertices adjacent to the current vertex (starting with the source). Update the columns corresponding to its neighbors with the direct edge weights from the source (if smaller than the existing value).
4. From the newest row, choose the vertex with the smallest value that has not yet been permanently marked. Call this vertex s_1.
5. Generate the next row by examining all neighbors of s_1. For each neighbor, update its column with the smaller of:
   (a) the sum of the smallest value from the previous row (distance to s_1) and the edge weight connecting s_1 to that neighbor;
   (b) the existing entry in that column.
6. Repeat the steps 4 and 5 until shortest distance from the destination vertex is determined or until all vertices are permanently marked.

Example

Consider a weighted graph with vertices A, B, C, D, E and edges:

$$A \to B : 4,\ \ A \to C : 2,\ \ B \to C : 3,\ \ B \to D : 2,\ \ C \to B : 1,\ \ C \to D : 4,\ \ D \to E : 2,\ \ E \to A : 7$$

We find the shortest distance from A to all other vertices.

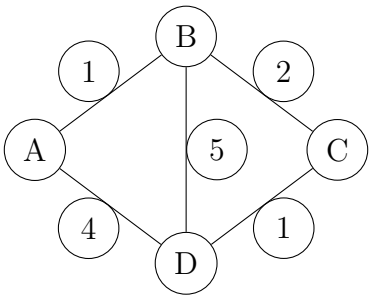
Step-by-step table:

Iteration	A	B	C	D	E
0	0	∞	∞	∞	∞
1 (from A)	0	4	2	∞	∞
2 (from C)	0	3	2	6	∞
3 (from B)	0	3	2	5	∞
4 (from D)	0	3	2	5	7

Final shortest distances from A:

$$A : 0, \quad B : 3, \quad C : 2, \quad D : 5, \quad E : 7$$

Exercise:



Find shortest path from A to all other vertices.

9.2 Travelling Salesman Problem (TSP)

Definition: The TSP is the problem of finding a Hamiltonian cycle (visiting each vertex exactly once and returning to the starting point) of minimal total cost in a weighted graph.

Model:

Given a list of cities and the distances between each pair of cities, find the shortest possible route that visits each city exactly once and returns to the origin city.

Algorithm Steps

1. List all vertices of the given weighted graph.
2. Choose a starting vertex (with maximum degree) (this will also be the ending vertex at the end of the tour).
3. From the current vertex, look at all edges leading to vertices not yet visited.
4. Select the edge with the smallest weight among them.
5. Move to that vertex and mark it as visited.
6. Repeat steps 3–5 until all vertices have been visited exactly once.
7. Return to the starting vertex by choosing the edge from the last visited vertex back to the starting vertex.
8. Sum up all the chosen edge weights to find the total travel cost.

Example

Consider a complete weighted graph G with vertices A, B, C, D and the following distance matrix:

	A	B	C	D
A	0	10	15	20
B	10	0	35	25
C	15	35	0	30
D	20	25	30	0

Solution

- Start at A.
- Nearest to A is B (distance 10). Path so far: $A \to B$.
- From B, nearest unvisited vertex is D (distance 25). Path: $A \to B \to D$.
- From D, nearest unvisited vertex is C (distance 30). Path: $A \to B \to D \to C$.
- From C, return to A (distance 15). Final tour: $A \to B \to D \to C \to A$.

$$\text{Total distance} = 10 + 25 + 30 + 15 = 80$$

Final Answer

Optimal tour found by nearest neighbour:

$$A \to B \to D \to C \to A$$

with total distance 80.

9.3 Chinese Postman Problem (CPP)

Definition: The CPP (or Route Inspection Problem) asks: What is the shortest closed walk that visits every edge of a graph at least once?

When is a solution optimal?

If a graph is Eulerian (all vertices have even degree), then an Euler circuit gives an optimal solution.

Simple Step-by-Step Algorithm (Undirected Case)

1. **Check connectivity.** Ensure G is connected. (Otherwise, there is no single closed route covering all edges.)
2. **Compute degrees.** For each vertex $v \in V$, compute its degree $\deg(v)$.
3. **If all degrees are even:** G is Eulerian.
   An Eulerian circuit exists; the minimum route length is $\sum_{e \in E} w(e)$, and any Eulerian circuit is optimal.
4. **If there are odd-degree vertices:**

(a) Write down which vertices have an odd degree (an odd number of edges connected to them).
Note: The number of odd-degree vertices will always be even.
(b) Find the shortest distances between every pair of these odd-degree vertices. (Here, distance means the length of the shortest path in the graph, not just a single edge.)
(c) Pair up the odd-degree vertices in such a way that the total extra distance is as small as possible. (This just means: connect odd vertices in pairs, choosing the cheapest ways to connect them.)
(d) Add these extra connections to the graph. (Think of it as duplicating some edges or paths so that all vertices now have even degree.)
(e) Once all vertices have even degree, find an Eulerian circuit (a path that covers every edge exactly once and returns to the start). This circuit will give the shortest possible route for the postman.

5. **Find an Eulerian circuit** in the augmented multigraph (original edges + duplicated edges), e.g., by Hierholzer's algorithm. This circuit is the optimal Chinese Postman tour.
6. **Cost.** The optimal tour length equals

$$\sum_{e \in E} w(e) \; + \; (\text{weight of the minimum perfect matching on } O).$$

Worked Example

Graph

Vertices A, B, C, D, E with edge weights:

$$A{-}B : 3, \quad A{-}C : 4, \quad B{-}C : 2, \quad B{-}D : 6, \quad C{-}D : 3, \quad C{-}E : 5, \quad D{-}E : 4.$$

Total weight of all edges:

$$3 + 4 + 2 + 6 + 3 + 5 + 4 = 27.$$

Degrees & Odd Vertices

$$\deg(A) = 2,\ \deg(B) = 3,\ \deg(C) = 4,\ \deg(D) = 3,\ \deg(E) = 2.$$

Odd-degree vertices: $O = \{B, D\}$.

Shortest Path Between Odd Vertices

We only need the distance between B and D:

$$\text{Direct } B{-}D = 6, \qquad B \to C \to D = 2 + 3 = 5.$$

Thus the shortest B–D path has length 5 (via $B{-}C{-}D$).

Augmentation (Make All Degrees Even)

Match the odd set O by the only possible pair $\{B, D\}$ and duplicate one shortest path between B and D, namely $B{-}C$ and $C{-}D$. This makes all degrees even.

Optimal Cost

$$\text{CPP cost} = \underbrace{27}_{\text{sum of all edges}} + \underbrace{5}_{\text{added shortest path}} = 32.$$

An Eulerian Circuit (One Valid Route)

In the augmented multigraph (with $B-C$ and $C-D$ duplicated), one Eulerian circuit is:

$$A \to B \to C \to D \to E \to C \to D \to B \to C \to A.$$

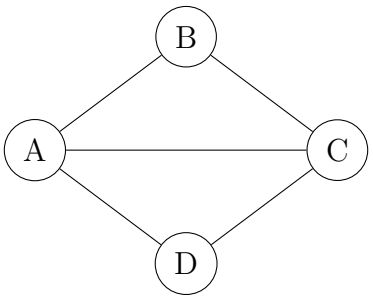
This closed walk uses every edge at least once and has total length 32.

Notes.

- If all degrees are even at the start, no duplication is needed.
- When more than two vertices are odd, compute all-pairs shortest paths among odd vertices, then choose a minimum-weight perfect matching on them (for small cases this can be done by inspection; in general use matching algorithms).

Example:

Consider a graph:



Degrees: A (3), B (2), C (3), D (2) $\Rightarrow$ A and C are odd. Add edge A–C again. Then find Euler circuit.